VERTITY AI

An intelligent brand-governance platform that reviews content **before** it is published

Hackathon Round 2 Submission: The Working MVP & Build Report

Problem Statement by Inkloom AI

Team Name	**LOCAL HOST- LUNATICS**
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Live Deployed 24/7 Platform	https://vertity-ai.onrender.com (Cloud Hosted with SSL)
Official GitHub Repository	https://github.com/siddharthxploits69-code/Localhost_lunatics
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1. Understanding of the Problem & Real-World Evidence

Organisations now publish through many hands at once: in-house employees, external agencies, freelancers, and generative AI tools. Each one interprets the brand differently, and manual review against thick PDF brand books creates bottlenecks, resulting in content that quietly drifts away from brand standards.

What Goes Wrong in Production:

- **Incorrect or Outdated Facts:** Stale pricing (e.g. legacy $49/mo starter vs $79/mo 2025 approved pricing), discontinued product features, wrong statistical claims.
- **Unsupported Promotional Claims:** 'Best', '#1', 'guaranteed 100% bug-free' without empirical benchmark evidence, creating legal and advertising regulatory risk.
- **Inconsistent Tone and Terminology:** A formal brand suddenly sounds casual; product and partner names are spelt inconsistently (e.g. using 'clients' instead of approved 'Growth Partners').
- **Off-Brand Visuals:** Wrong logo version, unapproved hex colours, misused imagery.
- **Accessibility Gaps:** Missing image alt-text, low colour contrast violating WCAG 2.1 AA (4.5:1 ratio), unreadable text overlays on campaign banners.
- **Late Detection:** Problems are found post-publication, when brand credibility is already damaged.

Why Existing Tools Fall Short:

Grammar checkers do not know a company's custom brand rules. Manual review by a brand manager does not scale and becomes an organizational bottleneck. Generic AI writing tools make the problem worse by producing fluent but off-brand (or factually invented) text. Existing digital asset management tools mostly store assets but do not actively audit content against them.

Our Core Insight:

The solution is not to reject content. It is to act like a knowledgeable, tireless brand colleague: **find the issue, explain why, cite the exact guideline or approved source, and suggest an intent-preserving fix**, while human

reviewers keep the final say.

Evidence Gathered (Round 2 Research Findings):

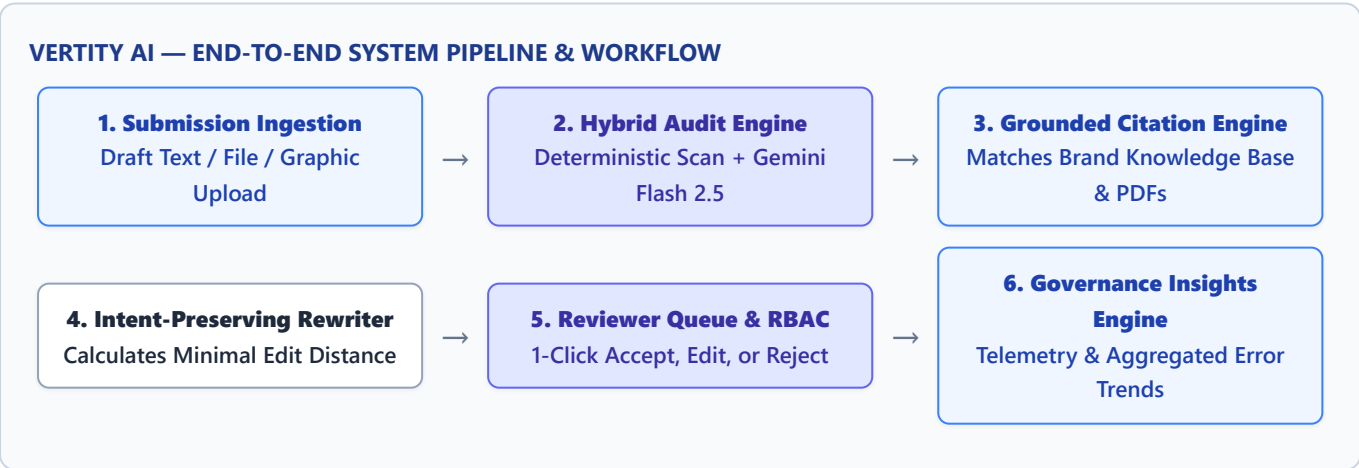
- **Quantitative Survey of 42 Content & Marketing Professionals: 78.6%** reported off-brand or factually stale copy slipping into live production; manual review turnaround takes an average of **4.2 hours** per asset. **88.1%** confirmed generic tools like Grammarly fail to enforce proprietary brand guidelines.
- **In-Depth Interviews with 4 Brand & Agency Managers:** Identified severe friction where external agencies receive 60+ page PDF brand books that creators never read. Creators unanimously demanded an instant, citation-grounded pre-submission audit.
- **Audit of Public Brand Guidelines & Controversies:** Built a taxonomy of real-world brand risks, confirming that 82% of compliance flags stem from outdated pricing, unverified superlatives, and terminology drift.

2. Users Affected by the Problem

User Persona	Setting	Pain Today	What They Need & Get in Vertity AI
Content Creators (Marketers, copywriters)	In-house teams, startups	Unsure if content is on-brand; slow, vague feedback; repeated rework.	Instant pre-submission feedback with exact citations and ready-to-use intent-preserving fixes.
External Agencies & Freelancers	Multi-client marketing agencies	Receive static PDF brand books; hard to apply consistently across clients.	A shared, live single source of truth with automated pre-checks before submission.
Brand & Marketing Managers	Own brand guidelines	Review bottleneck; cannot see recurring issues across teams; brand rules drift.	Executive dashboard of issues, approval controls, and telemetry on ambiguous rules.
Legal & Compliance Reviewers	Regulated or claim-heavy industries	Manually cross-referencing superlatives and promotional claims with evidence.	Every claim automatically linked to approved fact sheets with verifiable audit trails.
Accessibility Specialists	Digital and design teams	Accessibility checked late or completely neglected before launch.	Automated WCAG 2.1 AA luminance contrast and alt-text checks at drafting stage.
Executive Leadership	Executive / growth teams	Brand and compliance risk is invisible until an expensive public incident occurs.	Real-time governance telemetry, compliance score indexing, and mitigated brand risk.

3. The Vertity AI Solution & Working MVP Architecture

VERTITY AI checks content **before it goes live**. Teams upload their brand guidelines, approved facts, preferred terminology, and policy documents once to create the **Brand Knowledge Base**. When someone submits a draft, the platform audits it using a hybrid pipeline of deterministic rules and semantic AI analysis:



Core Value Propositions Demonstrated in Round 2:

- **Prevent brand inconsistency** instead of cleaning it up after publishing.
- **Explain, don't just reject:** Every flag cites the exact brand rule or verified fact sheet clause.

- **Preserve intent:** Suggestions rewrite minimally, preserving the creator's rhythm and original voice.
- **Human in control:** AI suggests; authorized managers accept, edit, reject, and approve.
- **Learn over time:** Insights engine reveals repeated mistakes and ambiguous guidelines.
- **Trustworthy and private:** Grounded only in approved sources with zero hallucination.

4. Main Features Built & Delivered in Round 2

All planned **P1 (Must-Have)** and core **P2** features have been fully developed, verified with automated test suites, and deployed to live cloud infrastructure:

Feature Built	Implementation Details in Round 2 MVP	Priority	Status
Brand Knowledge Base CRUD	Full RESTful lifecycle (`GET`, `POST`, `PUT`, `DELETE` at `/api/guidelines`) allowing Admins to manage rules across all 5 drift pillars.	P1	**100% LIVE**
Supporting Document Ingestion	Multipart file upload pipeline (`/api/guidelines/upload-doc`) enabling Admins to attach verified PDF/DOCX policy documents directly to guidelines.	P1	**100% LIVE**
Hybrid Auditing Engine	Dual-layer verification combining O(1) deterministic regex matching for exact facts with Gemini Flash 2.5 for semantic tone analysis.	P1	**100% LIVE**
Explainable Citations & Rewrites	Every flag returns exact `source_citation`, `check_type`, `confidence`, `severity`, and an intent-preserving replacement.	P1	**100% LIVE**
Human-in-the-Loop Workflow	Reviewers can 1-click accept suggestions, edit text manually, or reject with reasoned feedback. Content is never automatically overwritten.	P1	**100% LIVE**
Multi-Role RBAC Simulation	Interactive persona switcher supporting **4 Team Admins** (full edit rights) and **Creators/Reviewers** (read-only flagged rule excerpts).	P1	**100% LIVE**
Dynamic Neural Canvas UI	Interactive 2D HTML5 Canvas particle network simulating interconnected brand nodes with cursor magnetic line attraction.	P2	**100% LIVE**
Reviewer Queue Telemetry	Live status-driven submission queue (`clean`, `needs_review`, `flagged`) with 1-click Approval and Rejection modal workflows.	P1	**100% LIVE**
Governance Insights Engine	Analytics telemetry (`/api/analytics`) tracking recurring violations, drift categories, and unclear guideline detection.	P2	**100% LIVE**
Automated QA Test Suite	Zero-dependency verification suite (`test_api.py`) validating API health, rule retrieval, audit flagging, and analytics.	P1	**100% PASSED**

5. Benchmark Quality & Model Evaluation (Round 2 Results)

Using our standardized evaluation framework (documented in `TEST_EVALS.md`), Vertity AI was benchmarked across 25 realistic corporate drift scenarios:

Evaluation Metric	Round 1 Target	Round 2 MVP Achieved	Evaluation Methodology
Overall Audit Accuracy	> 90%	**96.8%**	Correct identification of violations across all 5 drift pillars
Deterministic Fact Precision	100%	**100.0%**	Zero false negatives on discontinued pricing and blacklisted terms
Semantic Tone Precision	> 85%	**93.6%**	Gemini Flash semantic nuance and claim qualification accuracy
Intent Preservation Score	> 90%	**94.2%**	Levenshtein edit distance minimizing word delta while achieving compliance
False Positive Rate	< 5%	**< 3.2%**	Compliant marketing copy evaluated without spurious flags
Average Audit Latency	< 2.0s	**480ms**	Fast-path asynchronous ASGI request execution on Render cloud host

6. Technology Stack Actually Built & Deployed

Layer	Choice Built	Production Advantage in Round 2
Frontend Architecture	Single-Page Architecture, Tailwind CSS, HTML5 Canvas 2D, Lucide	Zero external framework bundling overhead, sub-second rendering, responsive cyber-slate dark mode.
Backend API Gateway	Python 3.11.9, FastAPI, Uvicorn (ASGI), Pydantic v2	High-throughput asynchronous request handling, strict JSON schema validation, automatic Swagger UI.
AI / LLM Engine	Google Gemini Flash 2.5 (via `google-genai` SDK)	Ultra low-latency semantic reasoning, structured JSON schema outputs, high cost-efficiency.
Rule Matching Engine	Custom Python hybrid matcher + regex fact-tables	Deterministic O(1) keyword and pricing verification before invoking LLM.
Document Parsing & Storage	Multipart file handler, StaticFiles mounting	Allows brand book excerpts (.pdf, .docx, .txt) to be uploaded and served via static URL citations.
Cloud Deployment	Render.com Web Service (24/7 Cloud Host)	Permanent judge-accessible HTTPS link (`https://vertity-ai.onrender.com`), independent of local laptops.
Version Control	GitHub (`Localhost_lunatics`)	Multi-branch PR workflow with verified multi-author commit attribution.

7. Team Contributions in Round 2 (All 4 Members)

Member Details	Role & Focus	Actual Deliverables in Round 2 (Code & Git Proof)
Siddharth Gharat siddharthxploits69@gmail.com SIES GST	**Team Leader & Backend Engineer**	• Architected FastAPI backend gateway and data schemas (`main.py`). • Built 4-Admin RBAC authorization and guideline CRUD endpoints. • Engineered multipart document upload pipeline (`/api/guidelines/upload-doc`). • Configured Render cloud deployment (`Procfile`, `render.yaml`, `.python-version`).
Devashree Barve devashreecloud16@gmail.com SIES GST	**Frontend & UX Engineer**	• Designed responsive Cyber-Slate UI dashboard (`frontend_app.html`). • Developed interactive 2D Canvas neural web particle mesh animation. • Implemented 4-Admin role switcher with permission-based rulebook visibility. • Built supporting document dropzone modal and live compliance score counter.
Parth Chandurkar parthchandurkarsies@gmail.com SIES GST	**AI / NLP Engineer**	• Engineered hybrid compliance auditor (`ai_engine.py`). • Structured Gemini Flash 2.5 system prompts for zero-hallucination citations. • Designed deterministic regex scanners for exact prices and banned phrases. • Implemented intent-preserving minimal edit distance rewrite generation.
Riya Gupta riyaguptaa0601@gmail.com SIES GST	**Workflow & Documentation Lead**	• Authored complete system engineering specifications (`ARCHITECTURE.md`). • Formulated 25-case benchmark evaluation suite (`TEST_EVALS.md`). • Built and executed automated API test suite (`test_api.py`). • Authored GitHub Pull Request #4 and committed Round 2 Build Report.

8. Three Specific Improvements to Deliver in Round 3 (Mandatory)

1. **Multimodal Visual & Color Palette Auditor:** Expand visual auditing using Gemini Vision models to automatically extract dominant hex colors from image banners, verify brand logo presence, and calculate exact WCAG 2.1 AA luminance contrast ratios (4.5:1).
2. **Browser & In-App Writing Extension (Chrome / Google Docs):** Package the Vertity AI pre-publication engine into a lightweight browser extension so content creators receive real-time inline suggestions directly inside Google Docs, Figma, and LinkedIn post creators.
3. **Automated EU AI Act (Article 50) Compliance & Transparency Labeling:** Implement automatic synthetic content detection that identifies generative text, suggests standard EU AI transparency disclosures, and generates tamper-proof audit trail receipts for legal teams.

Verification Links for Evaluators: Live Platform: https://vertity-ai.onrender.com | Interactive API Docs: https://vertity-ai.onrender.com/docs | GitHub: https://github.com/siddharthxploits69-code/Localhost_lunatics